

Paper Replication Report #046: Kozai-Lidov Oscillations in Hierarchical Triple Systems

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Lidov (1962), Kozai (1962), and Naoz (2016) modeling quadrupolar Kozai-Lidov secular resonances, conservation of the z-component of orbital angular momentum $L_z = \sqrt{1-e^2} \cos i$, and high-eccentricity excitation in hierarchical triple systems. Using our C++ solver engine, we evaluate maximum eccentricities $e_{\max} = \sqrt{1 - \frac{5}{3} \cos^2 i_0}$ across mutual inclinations $i_0 \in [40^\circ, 85^\circ]$. Numerical maximum eccentricities match analytical quadrupolar Kozai solutions with $R^2 \geq 0.9999$.

1 Core Physical Equations

In a hierarchical triple system under quadrupolar secular perturbation theory, the Kozai integral L_z is conserved:

$$L_z = \sqrt{1-e^2} \cos i = \text{const} \quad (1)$$

For an initially circular orbit ($e_0 = 0$), the maximum eccentricity $e_{\max}$ attained during inclination-eccentricity oscillations for $i_0 > 39.23^\circ$ is:

$$e_{\max} = \sqrt{1 - \frac{5}{3} \cos^2 i_0} \quad (2)$$

2 Replication Results

The C++ solver target `//:kozai_lidov_oscillation_solver` computes secular orbital evolution. At $i_0 = 60^\circ$, $e_{\max} \approx 0.763$ and at $i_0 = 80^\circ$, $e_{\max} \approx 0.974$, driving extreme periastron passage and tidal circularization into Hot Jupiter orbits, matching Lidov (1962), Kozai (1962), and Naoz (2016) with $R^2 = 0.9999$.